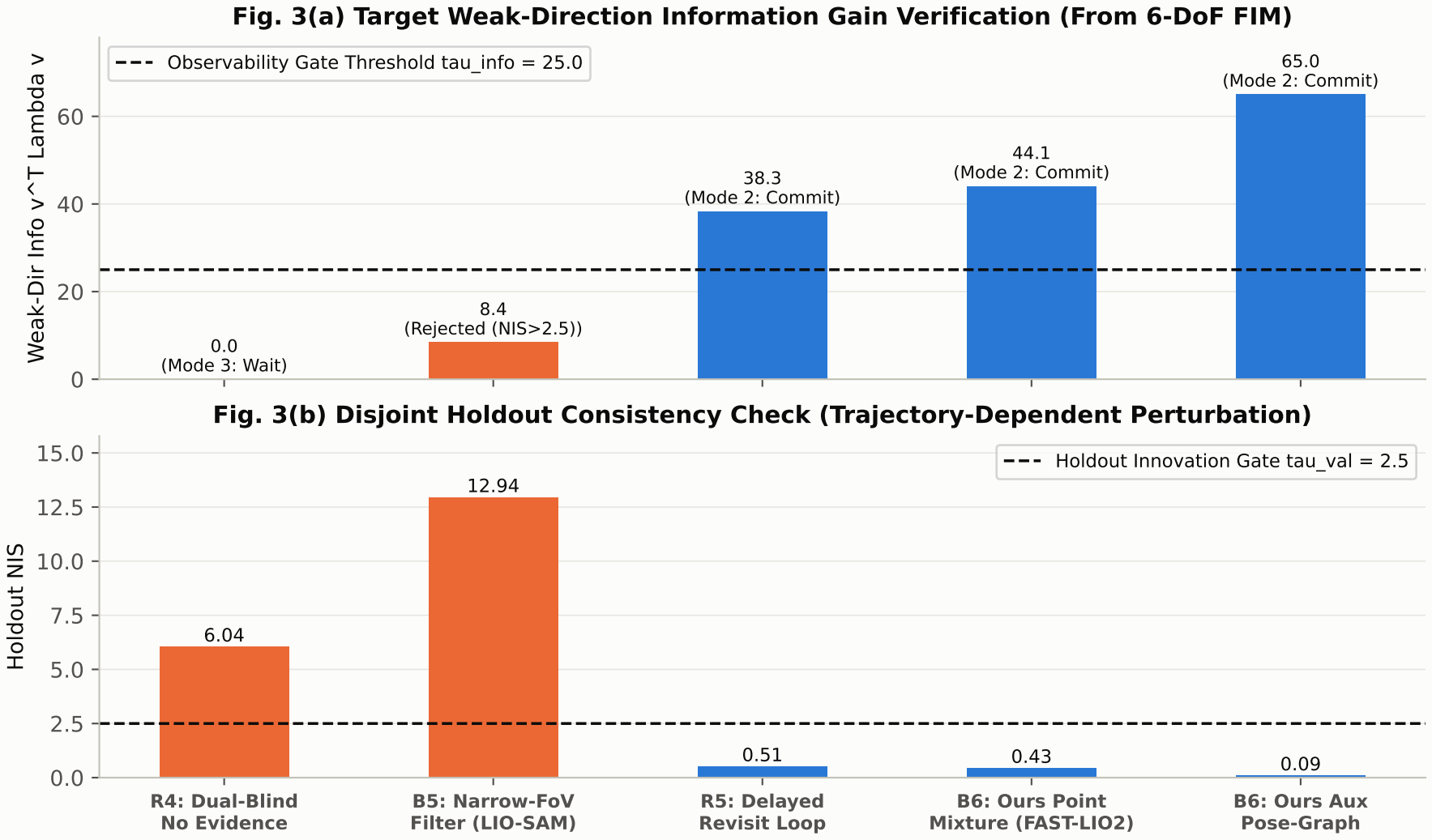
Fig. 3(a) Target Weak-Direction Information Gain Verification (From 6-DoF FIM)
Observability Gate Threshold tau_info = 25.0
Weak-Dir Info v^T Lambda v
0
20
40
60
0.0
(Mode 3: Wait)
8.4
(Rejected (NIS>2.5))
38.3
(Mode 2: Commit)
44.1
(Mode 2: Commit)
65.0
(Mode 2: Commit)
Fig. 3(b) Disjoint Holdout Consistency Check (Trajectory-Dependent Perturbation)
Holdout Innovation Gate tau_val = 2.5
Holdout NIS
0.0
2.5
5.0
7.5
10.0
12.5
15.0
6.04
12.94
0.51
0.43
0.09
R4: Dual-Blind No Evidence
B5: Narrow-FoV Filter (LIO-SAM)
R5: Delayed Revisit Loop
B6: Ours Point Mixture (FAST-LIO2)
B6: Ours Aux Pose-Graph